ANUJ KALE

Pune, Maharashtra | anujskale@gmail.com | Phone : +91 9168541230 | askyncme.github.io
Linkedin : www.linkedin.com/in/anuj-kale-672532291 | Github: https://github.com/askyncme

SUMMARY

Mechanical Engineering graduate with experience in autonomous UAVs, ROS2 robotics, computer vision, and embedded systems. Electronics Lead for Team Squadrone's SAE AeroTHON 2025 UAV project (AIR 11) and co-author of a published UAV research paper presented at DAUS 2026. Interested in navigation, guidance, control systems, and autonomous robotics.

EDUCATION

Bachelor of Engineering (BE) in Mechanical Engineering **July 2022 - July 2026**

Pune Vidyarthi Griha's College of Engineering and Technology -Savitribai Phule Pune University

- CGPA : 8.12
- HSC : 77%
- SSC : 90.4%

Honors in Robotics & Automation

SKILLS

- **UAV & Robotics:** ROS2, Gazebo, RViz, PX4, ArduPilot, Mission Planner, UAV System Architecture
- **Programming:** Python, OpenCV, YOLO, Embedded C, Basic C++
- **Navigation & Perception:** Sensor Integration, Computer Vision, Path Planning Fundamentals
- **Embedded Systems:** Arduino, ESP32, Raspberry Pi, Microcontroller Interfacing
- **Industrial Automation:** Mitsubishi FX3G PLC, Ladder Logic
- **Mechanical & Prototyping:** SolidWorks, 3D Printing

PROJECTS

1. **Team Squadrone - Autonomous UAV for Disaster Response**

SAE AeroTHON 2025 Phase I – AIR 11 | Research Paper Published at DAUS 2026, Salzburg, Austria

- Led electronics architecture and subsystem integration for an autonomous disaster-response UAV.
- Coordinated integration of flight controller, onboard computing, communication systems, sensors and mission payloads.
- Contributed to autonomous mission planning, target detection and payload deployment workflows using, Python and computer vision.
- Participated in UAV component selection, sensor integration, power system planning and simulation-based validation.
- Collaborated with multidisciplinary teams across electronics, mechanical and autonomous systems domains.
- Contributed to published research on terrain-aware target detection and autonomous payload delivery for UAV applications.

2. **Autonomous Mobile Assistant Robot (ROS2)**

- Developed a ROS2-based autonomous mobile robot using mecanum-drive kinematics.
- Implemented IMU and odometry integration for localization and navigation.
- Designed a navigation pipeline for autonomous indoor mobility.
- Utilized Gazebo and RViz for simulation, visualization and testing.

3. **Quadruped Robot**

- Led development of a 12-servo quadruped robot using Arduino Nano and 3D-printed components.
- Implemented gait control algorithms and servo coordination for stable locomotion.
- Conducted hardware integration, testing and performance validation.

4. **PLC-Based Pneumatic Gripper System**

- Developed a PLC-controlled pneumatic gripping system using Mitsubishi FX3G and a 5/3 solenoid valve.
- Programmed ladder logic for forward-reverse actuation and sensor-based control.

RESEARCH & PUBLICATIONS

Autonomous UAVs for Disaster Response: Terrain-Aware Target Detection and Payload Delivery

- Published research paper at the 2nd International Conference on Drones and Unmanned Systems (DAUS 2026), Salzburg, Austria.
- Investigated autonomous UAV systems for disaster-response missions, terrain-aware target detection and precision payload delivery.
- Explored integration of computer vision, onboard sensing and autonomous mission execution concepts.

EXPERIENCE

Procurement Intern, Unbox Robotics, Pune **Dec 2024 - Jan 2025**

- Supported technical procurement activities for warehouse robotics systems using SAP.
- Coordinated component sourcing, vendor communication and engineering support operations.
- Assisted in inventory tracking, documentation management and material planning.
- Gained exposure to robotics product-development workflows and cross-functional engineering operations.

ADDITIONAL INFORMATION

- **Languages :** English, Hindi, Marathi.
- **Extracurricular Interests :** Media Head (MESA & Samsara), EDC and TEDx design/media teams.